# Sequences

Quick reference sheet · GCSE Maths, Algebra · keep handy for revision

## 1. Term-to-term rules

A sequence is a list of numbers following a pattern; each number is a **term**. The **term-to-term rule** says how to get from one term to the next (e.g. "add 4", "subtract 3", "multiply by 2"). To continue a sequence, apply the rule to the last term you know.

3, 7, 11, 15, 19, ... → rule is **add 4**

50, 44, 38, 32, ... → rule is **subtract 6**

2, 6, 18, 54, ... → rule is **multiply by 3** (not every rule adds or subtracts)

## 2. Special sequences

| Name | Sequence | Pattern |
|---|---|---|
| Square numbers | 1, 4, 9, 16, 25, ... | n × n |
| Cube numbers | 1, 8, 27, 64, 125, ... | n × n × n |
| Triangular numbers | 1, 3, 6, 10, 15, ... | add 2, then 3, then 4, ... |
| Fibonacci-type | 1, 1, 2, 3, 5, 8, 13, ... | each term = sum of previous 2 |

## 3. Finding the nth term (arithmetic sequences)

The nth term is a formula that lets you jump straight to any term, without listing every term before it. It only works this simply when the sequence adds/subtracts the **same amount** each time.

**nth term = dn + (a – d) — d = common difference, a = first term**

### Method, step by step, for 3, 7, 11, 15, 19, ...

1. Find the common difference: d = 4 (each term goes up by 4)
2. Note the first term: a = 3
3. Work out a – d: 3 – 4 = –1
4. Write the nth term: **4n – 1**
5. Check it: substitute n = 1 → 4(1) – 1 = 3 ✓ (matches the first term)

With a negative (decreasing) difference: 50, 44, 38, 32, ...

d = –6, a = 50, a – d = 50 – (–6) = 56 → nth term = **–6n + 56**

## 4. Using the nth term

**To find a specific term, substitute the term number in for n.**

nth term = 3n + 4 → 10th term = 3(10) + 4 = **34**

**To check if a number is in the sequence, set the nth term formula equal to that number and solve for n.**

If n comes out as a **positive whole number**, it is in the sequence. If not (a decimal, a fraction, or negative), it is not.

Is 50 a term in the sequence with nth term 4n – 1?

$4n - 1 = 50 \rightarrow 4n = 51 \rightarrow n = 12.75 \rightarrow$ not a whole number $\rightarrow$ **No**

Is 47 a term in the sequence with nth term 3n + 2?

$3n + 2 = 47 \rightarrow 3n = 45 \rightarrow n = 15 \rightarrow$ whole number $\rightarrow$ **Yes** (it is the 15th term)